ATIKUL HOQUE

PhD Candidate in Geography and Environmental Studies
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RESEARCH INTERESTS

Remote sensing of dryland ecosystems · Shrub/woody plant encroachment · Deep learning for land cover mapping · Multi-sensor data fusion · Phenology · UAS/drone-based mapping · Rangeland monitoring · Google Earth Engine

EDUCATION

Doctor of Philosophy, Geography and Environmental Studies *2022 – Present*
New Mexico State University (NMSU), Las Cruces, NM

- Dissertation: Multi-sensor and deep learning approaches for mapping shrub encroachment in dryland ecosystems
- Passed PhD comprehensive examination; admitted to PhD candidacy
- GPA: 3.98 / 4.0
- Advisor: Dr. Michaela Buenemann

Master of Science, Natural Resource Management *2020 – 2022*
Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, The Netherlands

- Thesis: Variability of Wheat Phenology from Sentinel-1 and -2 Time Series: A Case Study for Brandenburg, Germany
- GPA: 7.95 / 10

Bachelor of Science, Geography and Environmental Studies *2015 – 2019*
University of Chittagong, Chattogram, Bangladesh

- GPA: 3.70 / 4.0

RESEARCH EXPERIENCE

Graduate Research Assistant *2022 – Present*
Spatial Application Research Center (SpARC), NMSU, Las Cruces, NM

- Conducting dissertation research on multi-sensor, deep learning-based mapping and monitoring of shrub encroachment in global drylands, with a focus on the Jornada Experimental Range, NM
- Developed a novel hybrid CNN-MLP framework for accurate classification of photosynthetic vegetation, non-photosynthetic vegetation, and bare soil in mesquite-encroached rangelands using UAS hyperspectral imagery
- Applied an attention-based fusion CNN integrating Sentinel-1 SAR and Sentinel-2 multispectral data for fractional cover mapping of shrubs at landscape scale
- Investigating optimal phenological windows for shrub mapping using multi-temporal satellite time-series data
- Led field campaigns to collect spectroradiometer measurements and UAS imagery as an FAA-certified remote pilot for radiometric calibration and validation

- Completed a comprehensive review of remote sensing-based methods for mapping shrub encroachment in global drylands
- Provided technical support (Google Earth Engine, R, Python, ENVI, ArcGIS/QGIS) to affiliated projects including land cover mapping in Panama and Acequia distribution prediction using GeoAI
- Presented research at national and regional conferences: AAG (2023, 2024, 2026), AGU (2025), SWAAG (2022, 2025), GRAS (2025)

Research Intern, German Remote Sensing Data Center *2022*

German Aerospace Center (DLR), Munich, Germany

- Developed a Google Earth Engine workflow to extract temporal profiles of Sentinel-2 EVI and vegetation indices across phenological phases of winter wheat
- Analyzed spatial and temporal variability of winter wheat phenology using SAR and optical time series alongside gridded weather and soil data
- Processed field parcel data in Python and QGIS for input into crop-type classification pipelines

Teaching and Research Assistant *2019 – 2020*

Department of Geography and Environmental Studies, University of Chittagong, Bangladesh

- Served as instructor of record for the undergraduate course 'Remote Sensing and Satellite Imagery'
- Served as lab instructor for the undergraduate course 'Geographic Information System'
- Performed geospatial and climatic data acquisition, analysis, and visualization for departmental research projects

TEACHING EXPERIENCE

Instructor of Record — Maps and GIScience *Spring 2026 – Present*

Department of Geography and Environmental Studies, NMSU, Las Cruces, NM

- Teaching an undergraduate-level introductory course covering GIS theory and practical applications across disciplines
- Designing and delivering lectures, labs, and assessments using ArcGIS Online and ArcGIS Pro

Instructor of Record — Independent Study (Undergraduate) *Spring 2026 – Present*

Department of Geography and Environmental Studies, NMSU, Las Cruces, NM

- Supervising an undergraduate student in an independent study focused on GIS fundamentals and applications
- Providing one-on-one mentorship and guiding hands-on exercises with ArcGIS Online and ArcGIS Pro

Graduate Teaching Assistant *2023 – Present*

Department of Geography and Environmental Studies, NMSU, Las Cruces, NM

- Courses supported: Programming for GIS, Advanced Remote Sensing, Cartography and GIS, and others
- Graded assignments and projects, provided technical support, and assisted students with course-related software and analysis

PUBLICATIONS

Under Review

1. Hoque, A., & Buenemann, M. (under review). From pixels to landscapes: Advances in remote sensing-based mapping of shrub encroachment in global drylands. *(Review manuscript, to be submitted)*
2. Hoque, A., & Buenemann, M. (under review). A hybrid CNN-MLP approach for classifying photosynthetic vegetation, non-photosynthetic vegetation, and soil in mesquite-encroached rangelands. *(Research manuscript, to be submitted)*

3. Hoque, A., & Buenemann, M. (under review). Sentinel-1 and Sentinel-2 data fusion with an attention-based CNN for fractional cover mapping of shrubs in drylands. *(Research manuscript, to be submitted)*

Peer-Reviewed Journal Articles

4. Hoque, A., Mahmud, A., Rashid, K. J., & Biswas, R. N. (2026). Assessing spatial variability of Boro rice (Oryza sativa L.) phenology across different climatic zones in Bangladesh using Sentinel-1 and -2 time series data. *Agricultural Research,* 1–24.
5. Mahmud, A., Hoque, A., Tuure, J., Alakukku, L., Mäkelä, P. S., Wachiye, S., … & Heiskanen, J. (2026). Integrating field camera imagery for monitoring maize phenology, biophysical traits, and agroclimatic factors in smallholder farms. *Computers and Electronics in Agriculture,* 240, 111161.
6. Paul, A., Hoque, A., Mahmud, A., Esha, T. A., Rashid, K. J., Paul, B., … & Nath, T. K. (2024). Water, sanitation, and hygiene conditions and health outcomes: A comparative study between Rohingya refugees and local people in southeastern Bangladesh. *The Dhaka University Journal of Earth and Environmental Sciences,* 13(1), 47–58.
7. Rashid, K. J., Hoque, A., Esha, T. A., Rahman, M. A., & Paul, A. (2021). Spatiotemporal changes of vegetation and land surface temperature in the refugee camps and surrounding areas of Bangladesh after the Rohingya influx from Myanmar. *Environment, Development and Sustainability,* 23, 3562–3577. https://doi.org/10.1007/s10668-020-00733-x

Thesis

8. Hoque, A. (2022). Variability of wheat phenology from Sentinel-1 and -2 time series: A case study for Brandenburg, Germany. *Master's Thesis,* University of Twente, Enschede, The Netherlands.

CONFERENCE PRESENTATIONS

Oral Presentations

1. **Hoque, A., & Buenemann, M. (2026).** Hybrid deep learning for UAS-based shrubland mapping: Optimizing spatial resolution and spectral features. *American Association of Geographers (AAG) Annual Meeting, San Francisco, CA.* (Oral)
2. **Hoque, A., & Buenemann, M. (2025).** Finding the sweet spot: Spatial resolution and feature set effects on mapping mesquite-encroached rangelands. *Southwest Division, American Association of Geographers (SWAAG), Las Cruces, NM.* (Oral)
3. **Hoque, A., & Buenemann, M. (2025).** Spatial resolution effects on mapping mesquite-encroached rangelands. *Graduate Research and Arts Symposium (GRAS), NMSU, Las Cruces, NM.* (Oral)
4. **Hoque, A. (2024).** Mapping woody plant encroachment with a multi-sensor approach. *American Association of Geographers (AAG) Annual Meeting, Honolulu, HI.* (Oral)
5. **Hoque, A., & Buenemann, M. (2023).** Monitoring changes of rangeland vegetation with remote sensing. *American Association of Geographers (AAG) Annual Meeting, Denver, CO.* (Oral)
6. **Buenemann, M., Akrasi, O., & Hoque, A. (2023).** Use of multi-temporal optical, radar, and fused optical-radar images for mapping land cover changes in north-central Panama. *American Association of Geographers (AAG) Annual Meeting, Denver, CO.* (Oral)
7. **Hoque, A. (2023).** Mapping woody plant encroachment with remote sensing. *Data Science Symposium, NMSU, Las Cruces, NM.* (Oral)

Poster Presentations

8. **Hoque, A., & Buenemann, M. (2025).** Comparing fractional cover mapping of mesquite using machine learning approaches. *American Geophysical Union (AGU) Annual Meeting, New Orleans, LA.* (Poster)

9. **Hoque, A., Ransom, A., Buenemann, M., & Reichenborn, M. (2022).** Use of remote sensing for monitoring rangeland vegetation changes following herbicide treatment. *Joint Meeting, Southwest Division of the American Association of Geographers (SWAAG), Arkansas.* (Poster)

HONORS, AWARDS & SCHOLARSHIPS

Awards & Recognition

- First Prize, Advanced Research – Data Science Category, Graduate Research and Arts Symposium (GRAS), NMSU, 2025
- Best Teaching Assistant Award, Department of Geography and Environmental Studies, NMSU, 2024
- Jesse Rossbach Scholarship, New Mexico Geographic Information Council (NMGIC), 2024

Grants & Travel Funding

- College of Arts & Sciences Graduate Student Travel Grant, NMSU, 2023, 2025
- AAG–GTU Travel Grant, American Association of Geographers, 2023
- Department of Geography and Environmental Studies Travel Grant, NMSU, 2023, 2024
- SWAAG Student Summer Research Scholarship, 2023

Fellowships & Scholarships

- NMSU Graduate Success Scholarship, 2022, 2023, 2024, 2025, 2026
- Erasmus Mobility Exchange Fund (internship in Germany), University of Twente, 2021
- Orange Knowledge Programme (OKP) Scholarship, Dutch Government / Nuffic, 2020
- ITC Excellence Scholarship, Faculty ITC, University of Twente, The Netherlands, 2020
- National Science and Technology (NST) Award for Research Proposal, Government of Bangladesh, 2019

TECHNICAL SKILLS

Programming: Python (primary), R, JavaScript (Google Earth Engine API)

Machine / Deep Learning: CNN, MLP, hybrid CNN-MLP, attention-based fusion networks, Random Forest, SVM, MESMA

Remote Sensing Software: Google Earth Engine, ENVI, SNAP, eCognition, FRAGSTATS, TIMESAT

GIS Software: ArcGIS Pro, ArcGIS Online, QGIS

UAS / Field: FAA-certified remote pilot (sUAS), spectroradiometer operation, field data collection and calibration

SERVICE & LEADERSHIP

Professional & Academic Organizations

- Vice President, Bangladesh Student Association, NMSU — 2025–Present
- President, Graduate Geography and Environmental Studies Student Organization (GGESO), NMSU — 2024–2025
- Secretary, Kappa Xi Gamma / Theta Upsilon Honor Association, NMSU Chapter — 2022–Present
- Member, American Association of Geographers (AAG) — 2022–Present
- Member, American Geophysical Union (AGU) — 2023–Present
- FAA-Certified Remote Pilot, small unmanned aircraft systems (sUAS) — 2023–Present

Community & Outreach

- Co-founder, National Environmental Awareness and Restoration (NEAR) organization; organized workshops and poster competitions on environmental issues
- Organizer, 'GIS Day' events at University of Chittagong, 2016–2018
- Volunteer tutor for secondary school students preparing for competitive entrance examinations, Bangladesh
- Founding Member, Chittagong University Scientific Society, 2018–2020
- Member, Bangladesh Geographical Society, 2018–2020

CERTIFICATIONS

- Python Scripting for Geoprocessing Workflows, ESRI, 2024
- Creating Python Scripts for Raster Analysis, ESRI, 2024
- Geographic Information System Specialization, University of California, Davis, 2020
- GIS, Mapping, and Spatial Analysis Specialization, University of Toronto, 2020